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(54) **ANALYTE MONITORING SYSTEM AND METHODS**
(71) Applicant: **ABBOTT DIABETES CARE INC.**,
Alameda, CA (US)
(72) Inventors: **Martin J. Fennell**, Concord, CA (US);
Lei He, Moraga, CA (US); **Mark Kent Sloan**, Redwood City, CA (US)
(73) Assignee: **ABBOTT DIABETES CARE INC.**,
Alameda, CA (US)
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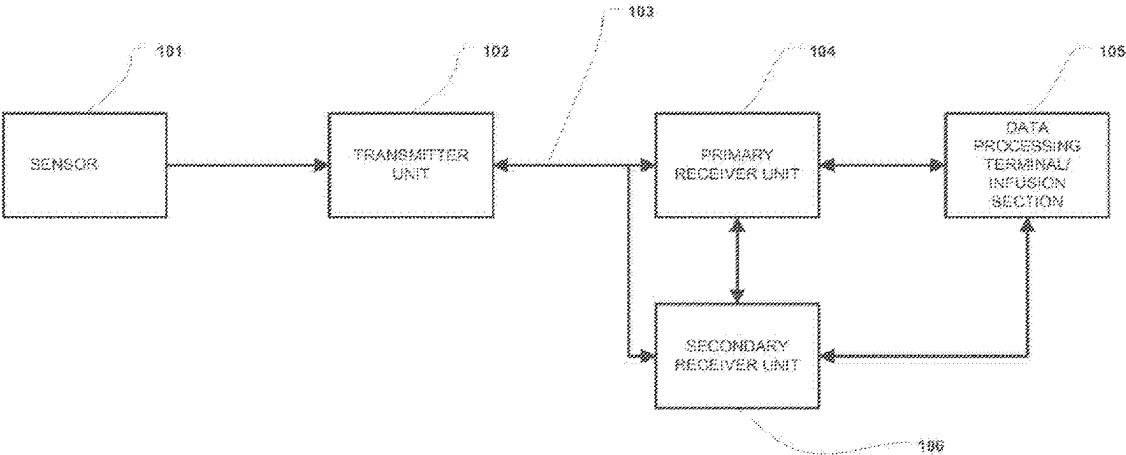
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Primary Examiner — Sisay Jacob
(74) *Attorney, Agent, or Firm* — Baker Botts L.L.P.

(57) **ABSTRACT**
Disclosed embodiments include methods and systems including a receiver unit of a glucose monitoring system. The receiver unit is configured to receive a key associated with a transmitter unit that enables the receiver unit to identify the transmitter unit, initiate communication with the transmitter unit, access a communication key uniquely associated with the transmitter unit, and receive communication packets from the remote transmitter unit on a periodic basis including data indicative of a glucose level of a bodily fluid. The receiver unit is further configured to process the data determine the glucose level for display, output a numerical representation of the determined glucose level in a GUI, and transmit data indicative of the glucose level to a second receiver unit.

28 Claims, 14 Drawing Sheets



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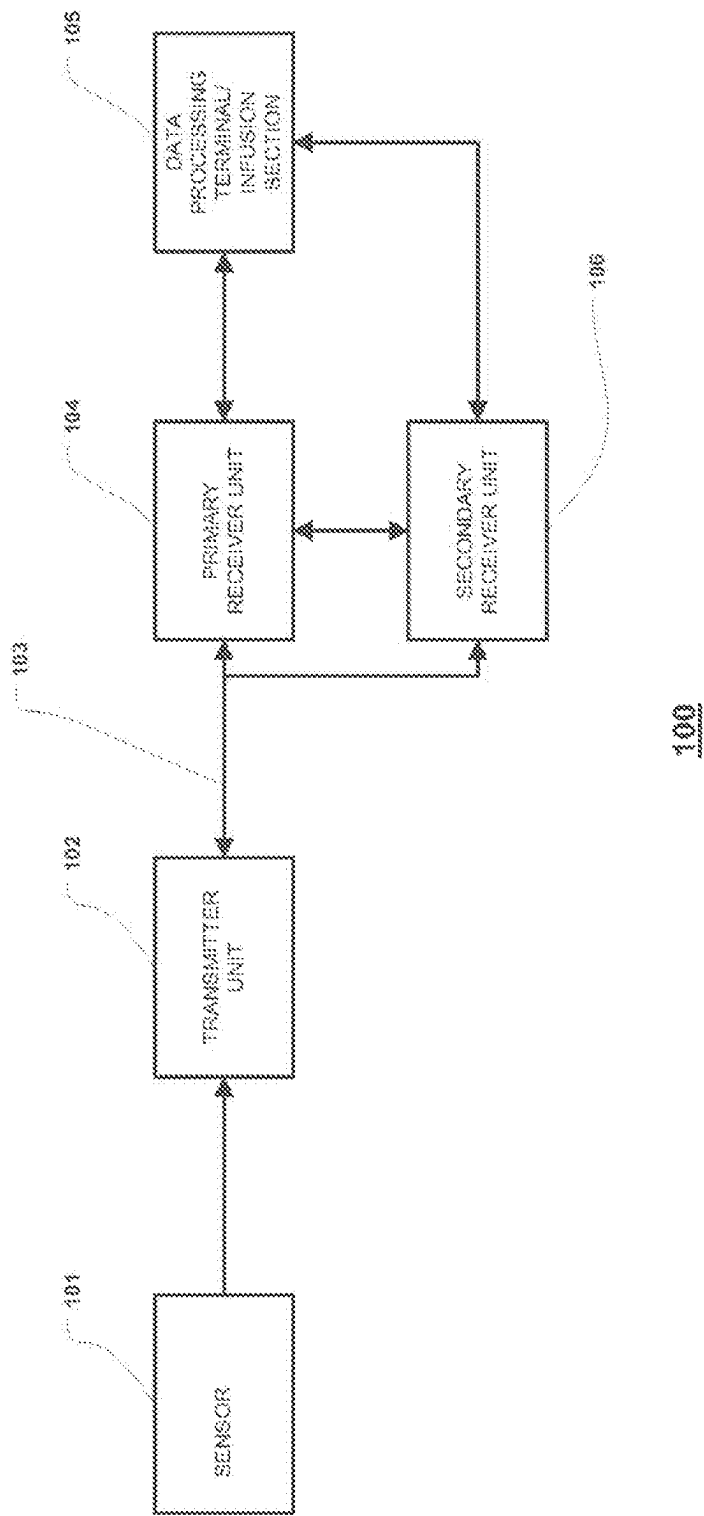


FIGURE 1

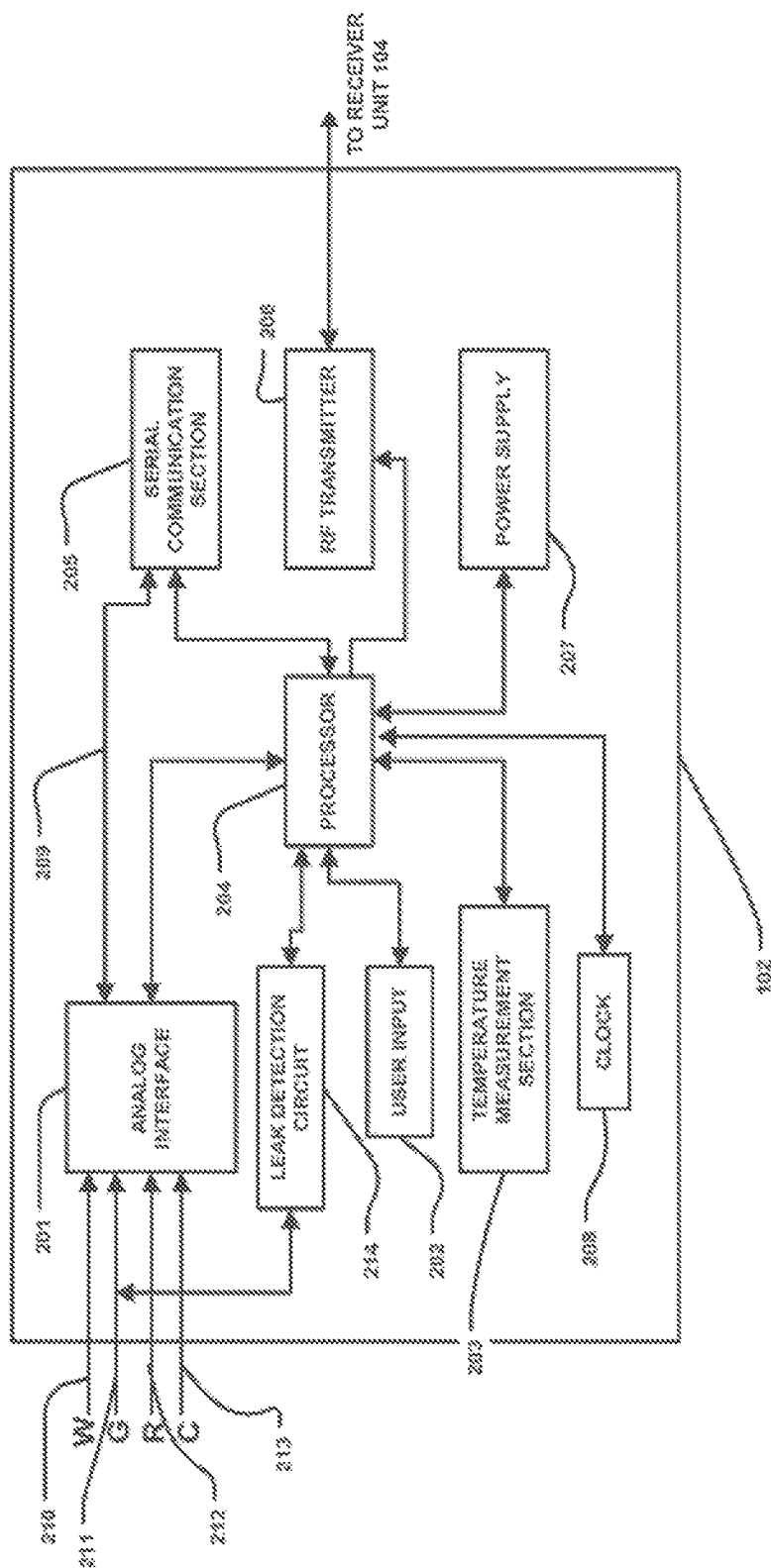


FIGURE 2

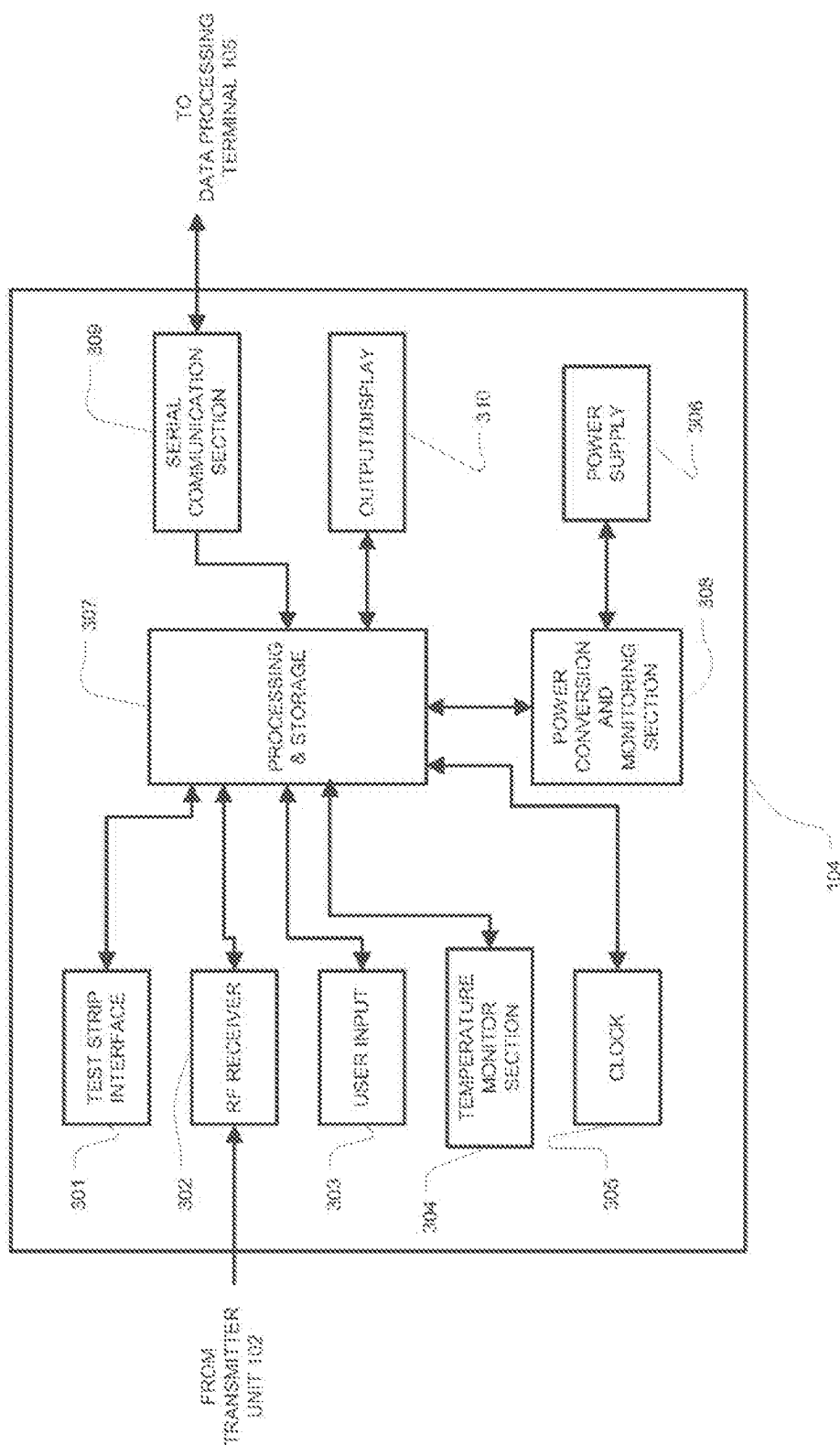


FIGURE 3

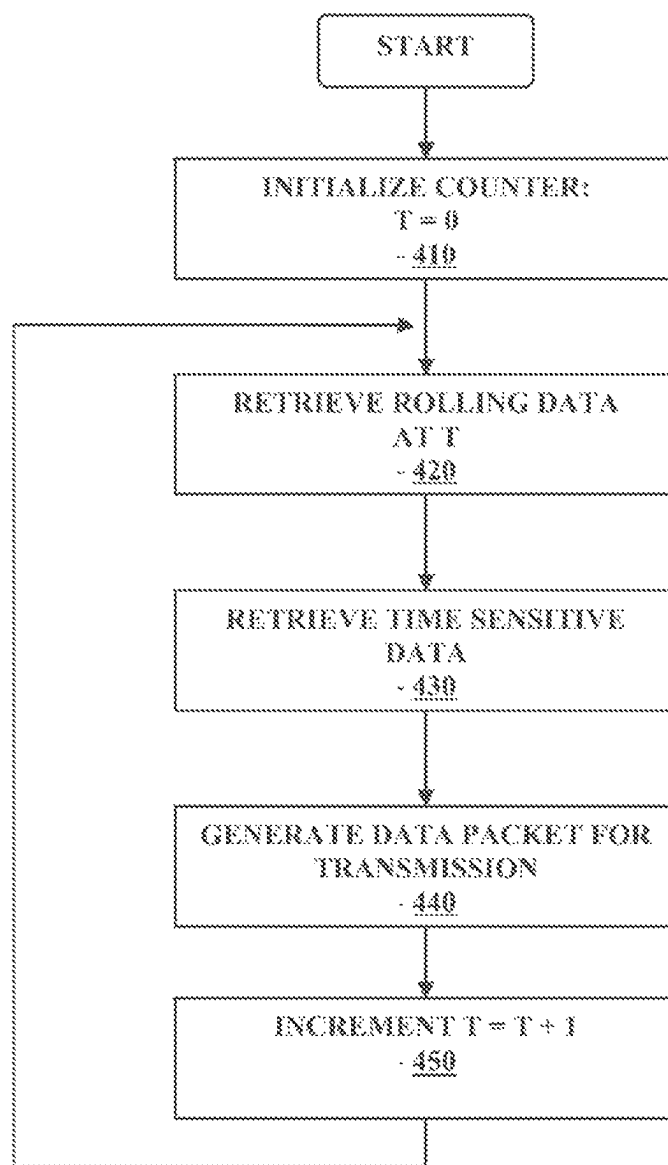


FIGURE 4

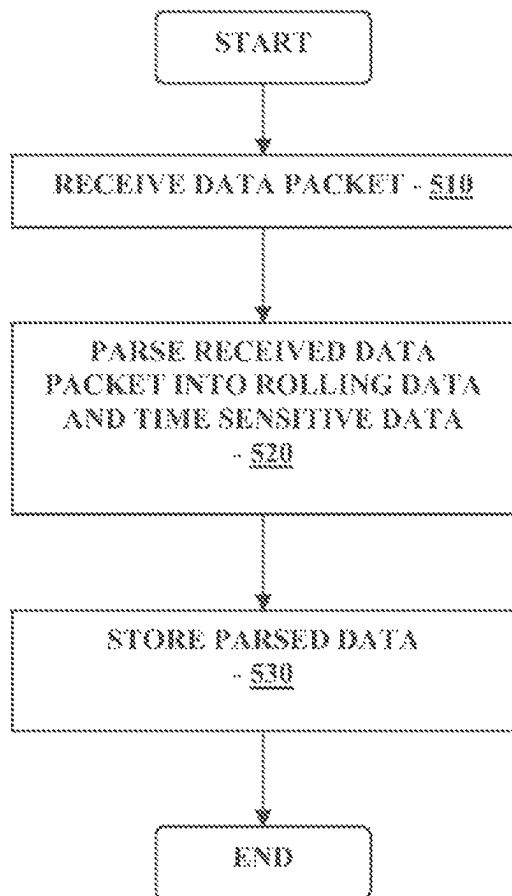


FIGURE 5

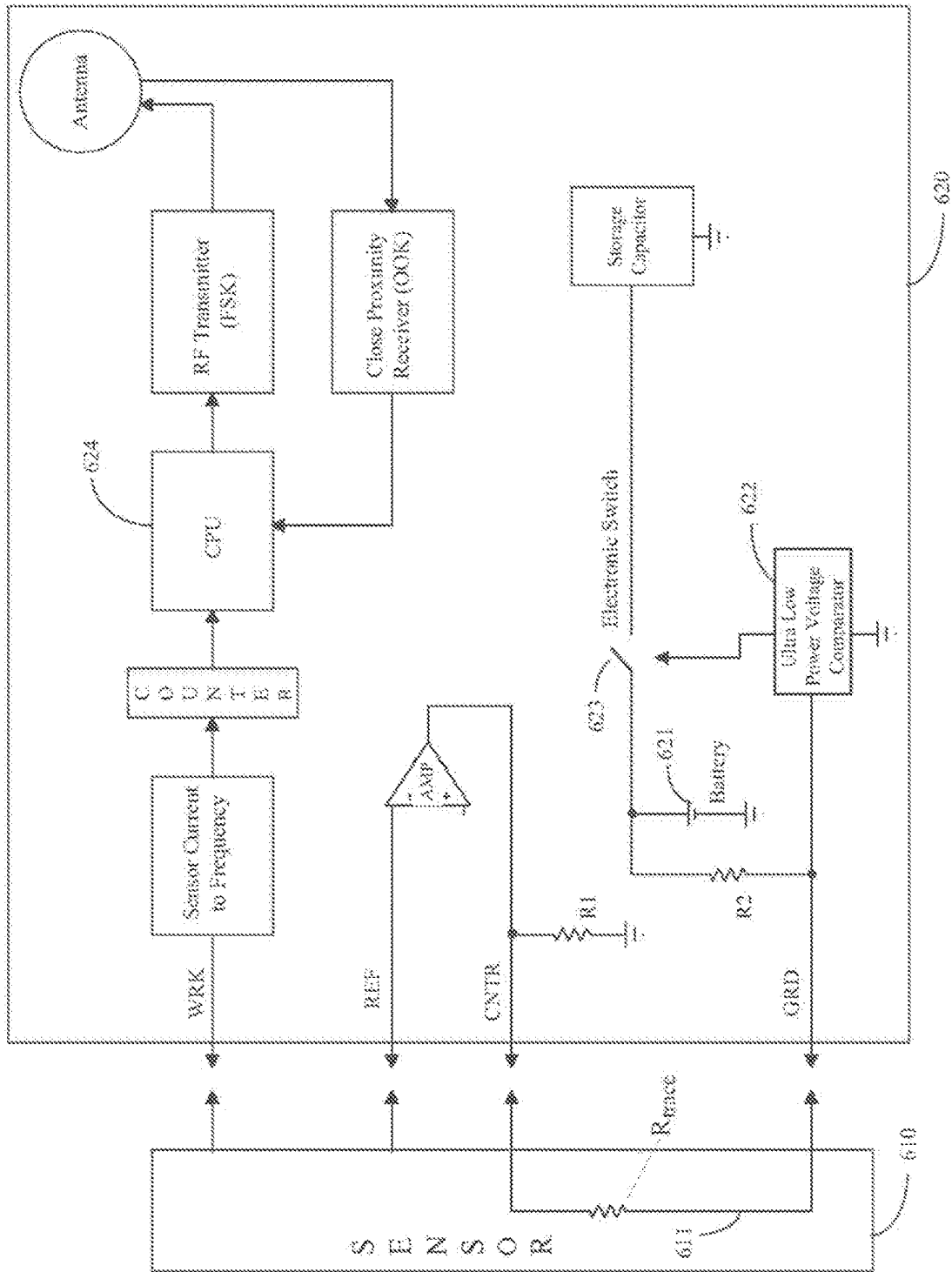


FIGURE 6

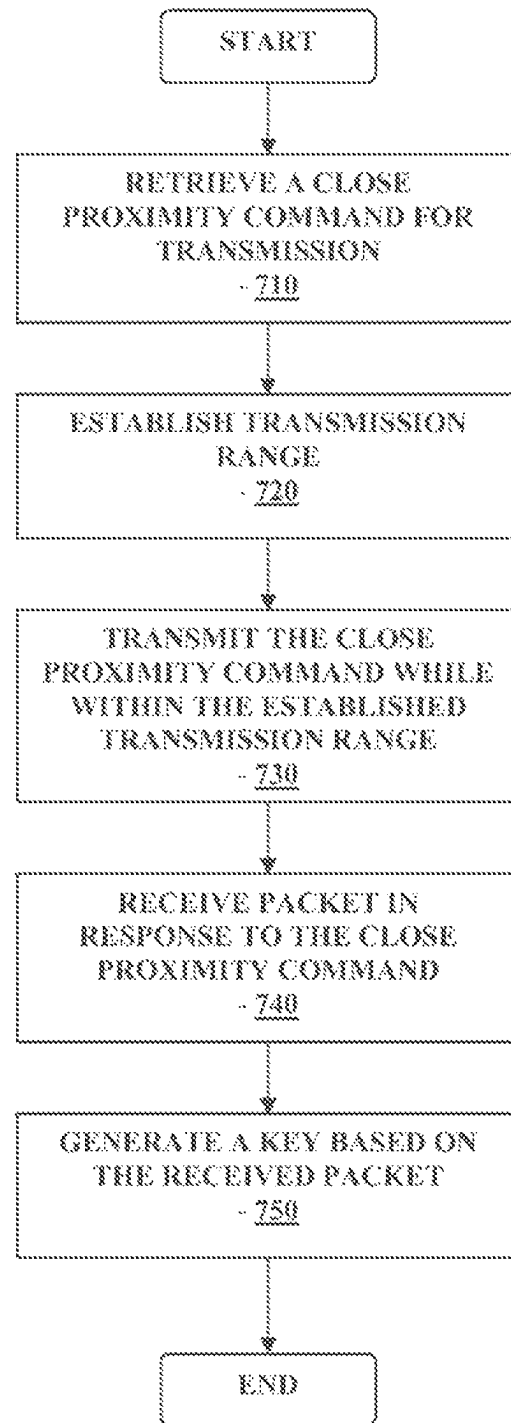


FIGURE 7

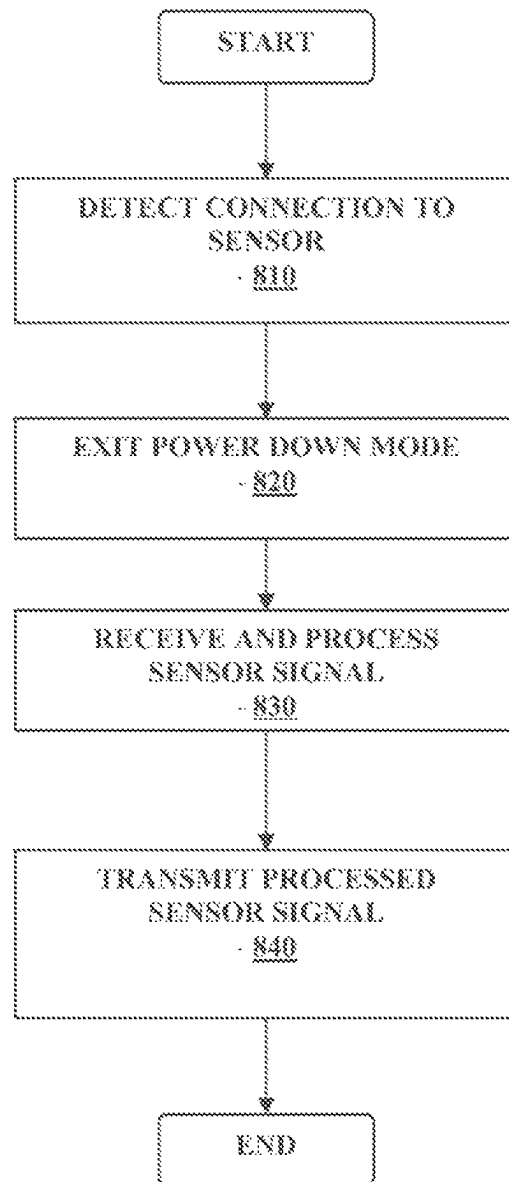


FIGURE 8

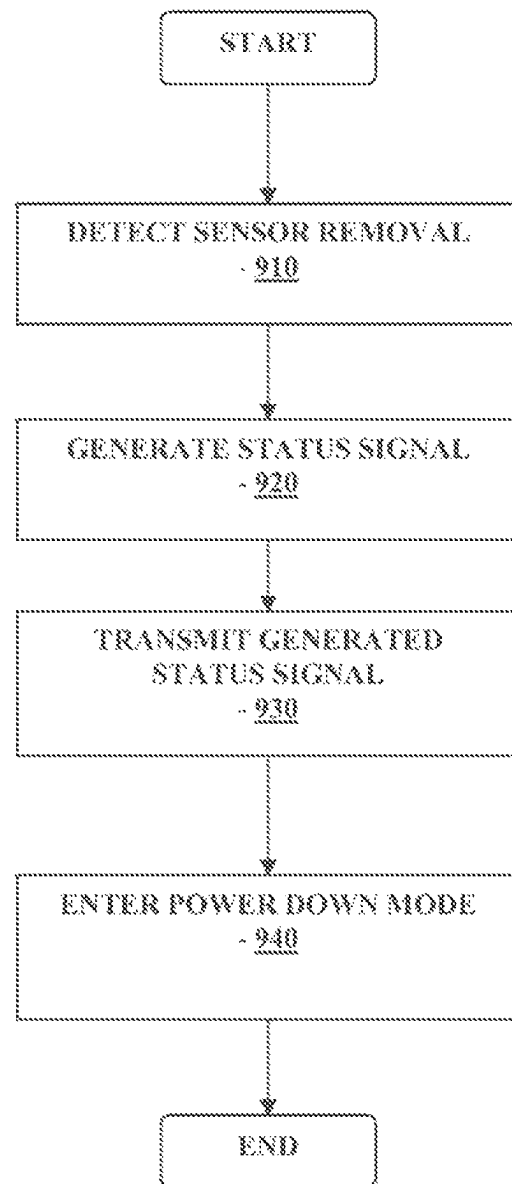


FIGURE 9

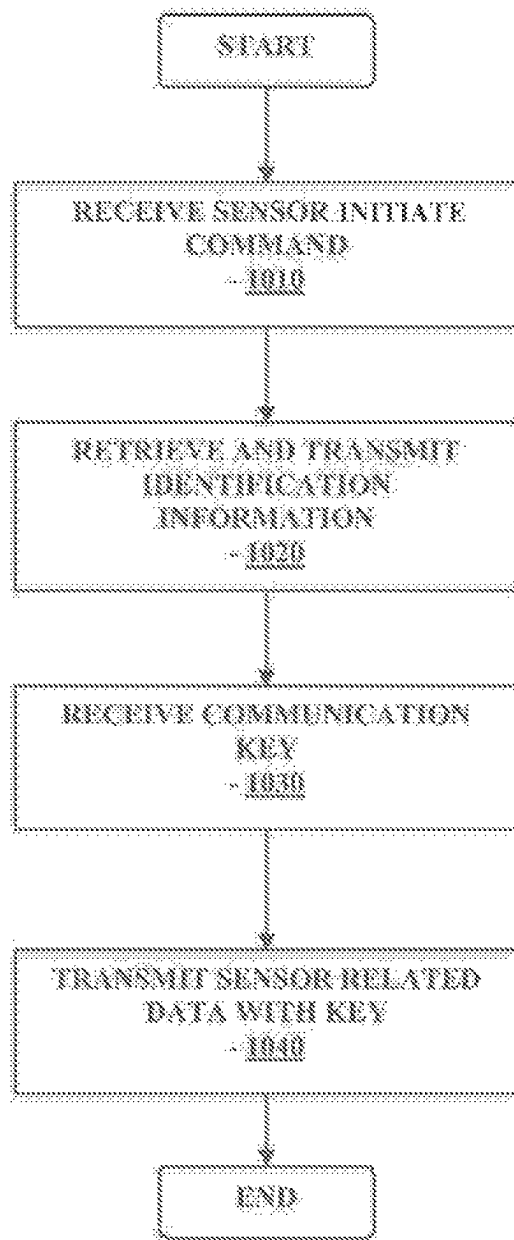


FIGURE 10

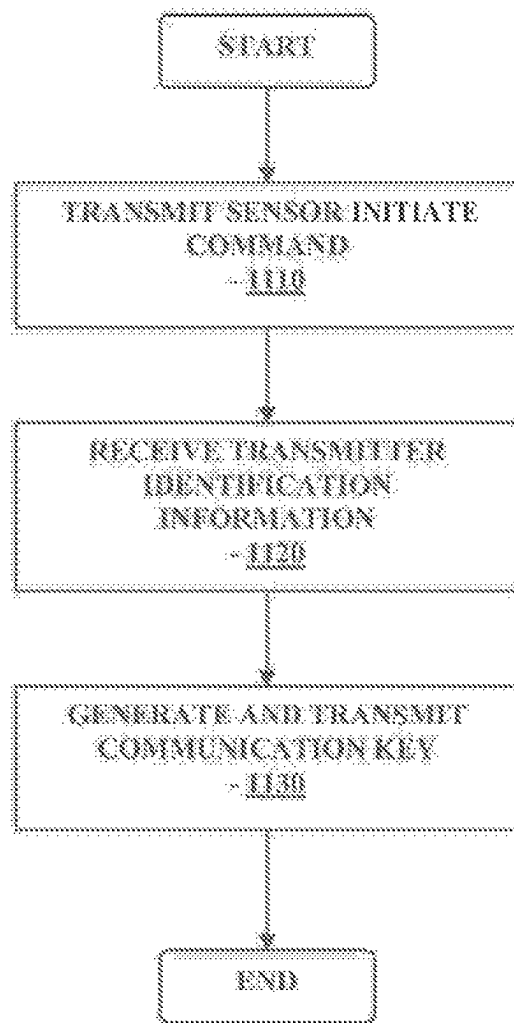


FIGURE 11

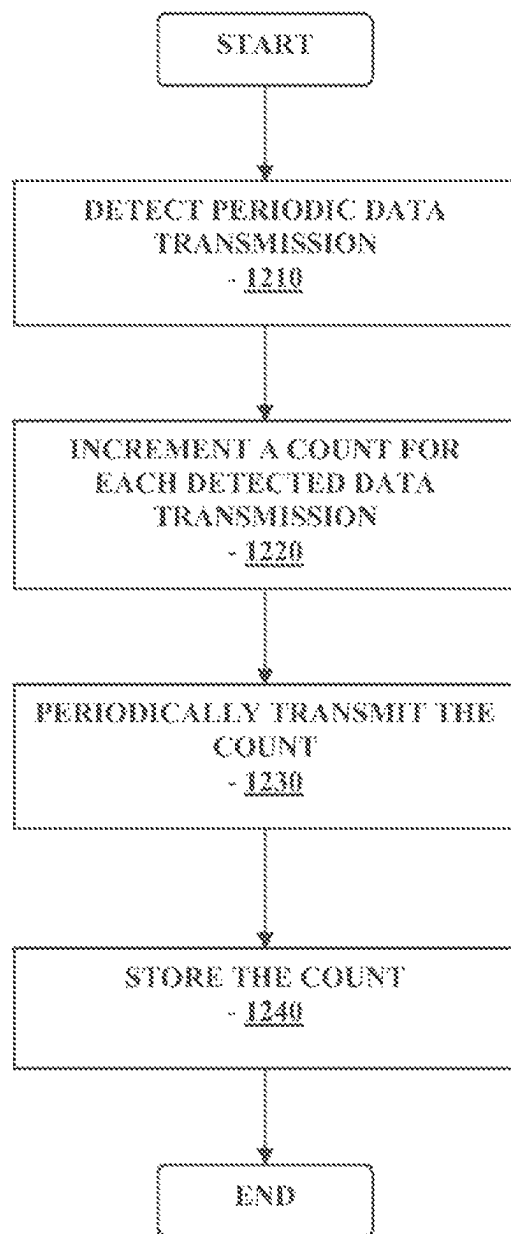


FIGURE 12

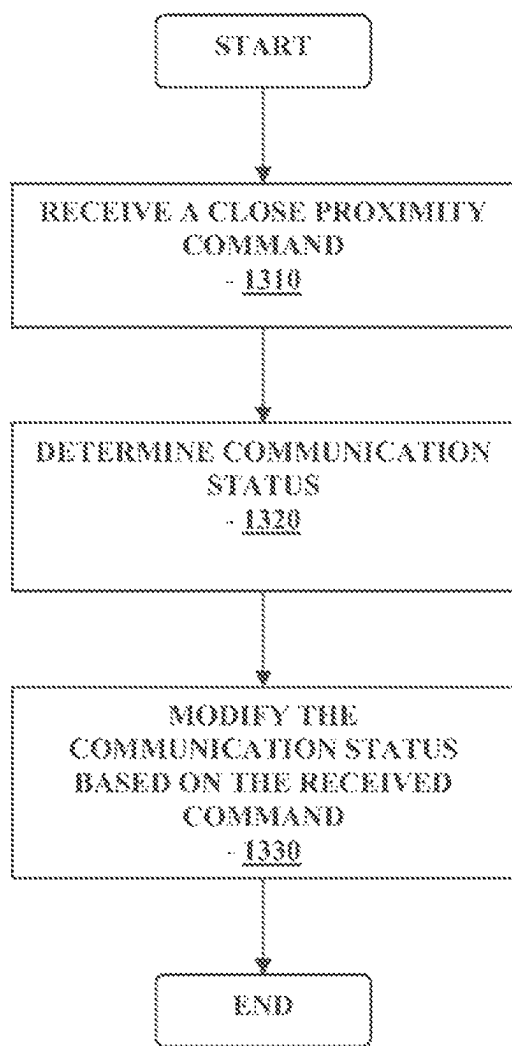


FIGURE 13

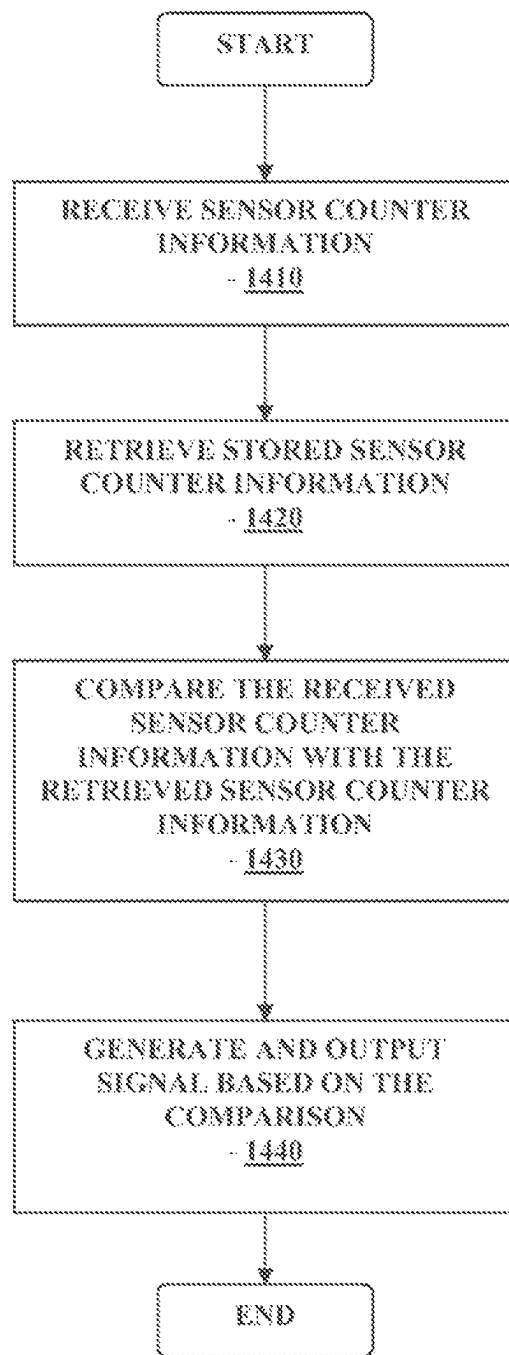


FIGURE 14

ANALYTE MONITORING SYSTEM AND METHODS

RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 17/874,616, filed Jul. 27, 2022, which is a continuation of U.S. patent application Ser. No. 17/179,589, filed Feb. 19, 2021, which is a continuation of U.S. patent application Ser. No. 16/850,943 filed Apr. 16, 2020, now U.S. Pat. No. 10,952,611, which is a continuation of U.S. patent application Ser. No. 16/245,160 filed Jan. 10, 2019, now U.S. Pat. No. 10,653,317, which is a continuation of U.S. patent application Ser. No. 15/591,073 filed May 9, 2017, now U.S. Pat. No. 10,178,954, which is a continuation of U.S. patent application Ser. No. 14/709,392 filed May 11, 2015, now U.S. Pat. No. 9,649,057, which is a continuation of U.S. patent application Ser. No. 13/906,288 filed May 30, 2013, now U.S. Pat. No. 9,035,767, which is a continuation of U.S. patent application Ser. No. 12/117,698 filed May 8, 2008, now U.S. Pat. No. 8,456,301, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 60/916,776 filed May 8, 2007, entitled "Analyte Monitoring System and Methods", the disclosures of each of which are incorporated herein by reference for all purposes.

BACKGROUND

Analyte, e.g., glucose monitoring systems including continuous and discrete monitoring systems generally include a small, lightweight battery powered and microprocessor controlled system which is configured to detect signals proportional to the corresponding measured glucose levels using an electrometer. RF signals may be used to transmit the collected data. One aspect of certain analyte monitoring systems include a transcutaneous or subcutaneous analyte sensor configuration which is, for example, at least partially positioned through the skin layer of a subject whose analyte level is to be monitored. The sensor may use a two or three-electrode (work, reference and counter electrodes) configuration driven by a controlled potential (potentiostat) analog circuit connected through a contact system.

An analyte sensor may be configured so that a portion thereof is placed under the skin of the patient so as to contact analyte of the patient, and another portion or segment of the analyte sensor may be in communication with the transmitter unit. The transmitter unit may be configured to transmit the analyte levels detected by the sensor over a wireless communication link such as an RF (radio frequency) communication link to a receiver/monitor unit. The receiver/monitor unit may perform data analysis, among other functions, on the received analyte levels to generate information pertaining to the monitored analyte levels.

Transmission of control or command data over wireless communication link is often constrained to occur within a substantially short time duration. In turn, the time constraint in data communication imposes limits on the type and size of data that may be transmitted during the transmission time period.

In view of the foregoing, it would be desirable to have a method and apparatus for optimizing the RF communication link between two or more communication devices, for example, in a medical communication system.

SUMMARY

Devices and methods for analyte monitoring, e.g., glucose monitoring, are provided. Embodiments include transmit-

ting information from a first location to a second, e.g., using a telemetry system such as RF telemetry. Systems herein include continuous analyte monitoring systems and discrete analyte monitoring system.

In one embodiment, a method including positioning a controller unit within a transmission range for close proximity communication, transmitting one or more predefined close proximity commands, and receiving a response packet in response to the transmitted one or more predefined close proximity commands, is disclosed, as well as devices and systems for the same.

These and other objects, features and advantages of the present invention will become more fully apparent from the following detailed description of the embodiments, the appended claims and the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a block diagram of a data monitoring and management system for practicing one or more embodiments of the present invention;

FIG. 2 is a block diagram of the transmitter unit of the data monitoring and management system shown in FIG. 1 in accordance with one embodiment of the present invention;

FIG. 3 is a block diagram of the receiver/monitor unit of the data monitoring and management system shown in FIG. 1 in accordance with one embodiment of the present invention;

FIG. 4 is a flowchart illustrating data packet procedure including rolling data for transmission in accordance with one embodiment of the present invention;

FIG. 5 is a flowchart illustrating data processing of the received data packet including the rolling data in accordance with one embodiment of the present invention;

FIG. 6 is a block diagram illustrating the sensor and the transmitter unit of the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention;

FIG. 7 is a flowchart illustrating data communication using close proximity commands in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention;

FIG. 8 is a flowchart illustrating sensor insertion detection routine in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention;

FIG. 9 is a flowchart illustrating sensor removal detection routine in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention;

FIG. 10 is a flowchart illustrating the pairing or synchronization routine in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention;

FIG. 11 is a flowchart illustrating the pairing or synchronization routine in the data monitoring and management system of FIG. 1 in accordance with another embodiment of the present invention;

FIG. 12 is a flowchart illustrating the power supply determination in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention;

FIG. 13 is a flowchart illustrating close proximity command for RF communication control in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention; and

FIG. 14 is a flowchart illustrating analyte sensor identification routine in accordance with one embodiment of the present invention.

DETAILED DESCRIPTION

As summarized above and as described in further detail below, in accordance with the various embodiments of the present invention, there is provided a method and system for positioning a controller unit within a transmission range for close proximity communication, transmitting one or more predefined close proximity commands, and receiving a response packet in response to the transmitted one or more predefined close proximity commands.

FIG. 1 illustrates a data monitoring and management system such as, for example, analyte (e.g., glucose) monitoring system 100 in accordance with one embodiment of the present invention. The subject invention is further described primarily with respect to a glucose monitoring system for convenience and such description is in no way intended to limit the scope of the invention. It is to be understood that the analyte monitoring system may be configured to monitor a variety of analytes, e.g., lactate, and the like.

Analytes that may be monitored include, for example, acetyl choline, amylase, bilirubin, cholesterol, chorionic gonadotropin, creatine kinase (e.g., CK-MB), creatine, DNA, fructosamine, glucose, glutamine, growth hormones, hormones, ketones, lactate, peroxide, prostate-specific antigen, prothrombin, RNA, thyroid stimulating hormone, and troponin. The concentration of drugs, such as, for example, antibiotics (e.g., gentamicin, vancomycin, and the like), digitoxin, digoxin, drugs of abuse, theophylline, and warfarin, may also be monitored. More than one analyte may be monitored by a single system, e.g. a single analyte sensor.

The analyte monitoring system 100 includes a sensor 101, a transmitter unit 102 coupleable to the sensor 101, and a primary receiver unit 104 which is configured to communicate with the transmitter unit 102 via a bi-directional communication link 103. The primary receiver unit 104 may be further configured to transmit data to a data processing terminal 105 in one embodiment may be configured to receive data directly from the transmitter unit 102 via a communication link which may optionally be configured for bi-directional communication. Accordingly, transmitter unit 102 and/or receiver unit 104 may include a transceiver.

Also shown in FIG. 1 is an optional secondary receiver unit 106 which is operatively coupled to the communication link and configured to receive data transmitted from the transmitter unit 102. Moreover, as shown in the Figure, the secondary receiver unit 106 is configured to communicate with the primary receiver unit 104 as well as the data processing terminal 105. Indeed, the secondary receiver unit 106 may be configured for bi-directional wireless communication with each or one of the primary receiver unit 104 and the data processing terminal 105. As discussed in further detail below, in one embodiment of the present invention, the secondary receiver unit 106 may be configured to include a limited number of functions and features as compared with the primary receiver unit 104. As such, the secondary receiver unit 106 may be configured substantially in a smaller compact housing or embodied in a device such as a wrist watch, pager, mobile phone, PDA, for example. Alternatively, the secondary receiver unit 106 may be configured with the same or substantially similar functionality

as the primary receiver unit 104. The receiver unit may be configured to be used in conjunction with a docking cradle unit, for example for one or more of the following or other functions: placement by bedside, for re-charging, for data management, for night time monitoring, and/or bi-directional communication device.

In one aspect, sensor 101 may include two or more sensors, each configured to communicate with transmitter unit 102. Furthermore, while only one transmitter unit 102, communication link 103, and data processing terminal 105 are shown in the embodiment of the analyte monitoring system 100 illustrated in FIG. 1, it will be appreciated by one of ordinary skill in the art that the analyte monitoring system 100 may include one or more sensors, multiple transmitter units 102, communication links 103, and data processing terminals 105. Moreover, within the scope of the present invention, the analyte monitoring system 100 may be a continuous monitoring system, or semi-continuous, or a discrete monitoring system. In a multi-component environment, each device is configured to be uniquely identified by each of the other devices in the system so that communication conflict is readily resolved between the various components within the analyte monitoring system 100.

In one embodiment of the present invention, the sensor 101 is physically positioned in or on the body of a user whose analyte level is being monitored. The sensor 101 may be configured to continuously sample the analyte level of the user and convert the sampled analyte level into a corresponding data signal for transmission by the transmitter unit 102. In certain embodiments, the transmitter unit 102 may be physically coupled to the sensor 101 so that both devices are integrated in a single housing and positioned on the user's body. The transmitter unit 102 may perform data processing such as filtering and encoding on data signals and/or other functions, each of which corresponds to a sampled analyte level of the user, and in any event transmitter unit 102 transmits analyte information to the primary receiver unit 104 via the communication link 103.

In one embodiment, the analyte monitoring system 100 is configured as a one-way RF communication path from the transmitter unit 102 to the primary receiver unit 104. In such embodiment, the transmitter unit 102 transmits the sampled data signals received from the sensor 101 without acknowledgement from the primary receiver unit 104 that the transmitted sampled data signals have been received. For example, the transmitter unit 102 may be configured to transmit the encoded sampled data signals at a fixed rate (e.g., at one minute intervals) after the completion of the initial power on procedure. Likewise, the primary receiver unit 104 may be configured to detect such transmitted encoded sampled data signals at predetermined time intervals. Alternatively, the analyte monitoring system 100 may be configured with a bi-directional RF (or otherwise) communication between the transmitter unit 102 and the primary receiver unit 104.

Additionally, in one aspect, the primary receiver unit 104 may include two sections. The first section is an analog interface section that is configured to communicate with the transmitter unit 102 via the communication link 103. In one embodiment, the analog interface section may include an RF receiver and an antenna for receiving and amplifying the data signals from the transmitter unit 102, which are thereafter, demodulated with a local oscillator and filtered through a band-pass filter. The second section of the primary receiver unit 104 is a data processing section which is configured to process the data signals received from the

5

transmitter unit **102** such as by performing data decoding, error detection and correction, data clock generation, and data bit recovery.

In operation, upon completing the power-on procedure, the primary receiver unit **104** is configured to detect the presence of the transmitter unit **102** within its range based on, for example, the strength of the detected data signals received from the transmitter unit **102** and/or predetermined transmitter identification information. Upon successful synchronization with the corresponding transmitter unit **102**, the primary receiver unit **104** is configured to begin receiving from the transmitter unit **102** data signals corresponding to the user's detected analyte level. More specifically, the primary receiver unit **104** in one embodiment is configured to perform synchronized time hopping with the corresponding synchronized transmitter unit **102** via the communication link **103** to obtain the user's detected analyte level.

Referring again to FIG. 1, the data processing terminal **105** may include a personal computer, a portable computer such as a laptop or a handheld device (e.g., personal digital assistants (PDAs)), and the like, each of which may be configured for data communication with the receiver via a wired or a wireless connection. Additionally, the data processing terminal **105** may further be connected to a data network (not shown) for storing, retrieving and updating data corresponding to the detected analyte level of the user.

Within the scope of the present invention, the data processing terminal **105** may include an infusion device such as an insulin infusion pump (external or implantable) or the like, which may be configured to administer insulin to patients, and which may be configured to communicate with the receiver unit **104** for receiving, among others, the measured analyte level. Alternatively, the receiver unit **104** may be configured to integrate or otherwise couple to an infusion device therein so that the receiver unit **104** is configured to administer insulin therapy to patients, for example, for administering and modifying basal profiles, as well as for determining appropriate boluses for administration based on, among others, the detected analyte levels received from the transmitter unit **102**.

Additionally, the transmitter unit **102**, the primary receiver unit **104** and the data processing terminal **105** may each be configured for bi-directional wireless communication such that each of the transmitter unit **102**, the primary receiver unit **104** and the data processing terminal **105** may be configured to communicate (that is, transmit data to and receive data from) with each other via the wireless communication link **103**. More specifically, the data processing terminal **105** may in one embodiment be configured to receive data directly from the transmitter unit **102** via a communication link, where the communication link, as described above, may be configured for bi-directional communication.

In this embodiment, the data processing terminal **105** which may include an insulin pump, may be configured to receive the analyte signals from the transmitter unit **102**, and thus, incorporate the functions of the receiver **104** including data processing for managing the patient's insulin therapy and analyte monitoring. In one embodiment, the communication link **103** may include one or more of an RF communication protocol, an infrared communication protocol, a Bluetooth® enabled communication protocol, an 802.11x wireless communication protocol, or an equivalent wireless communication protocol which would allow secure, wireless communication of several units (for example, per HIPAA requirements) while avoiding potential data collision and interference.

6

FIG. 2 is a block diagram of the transmitter of the data monitoring and detection system shown in FIG. 1 in accordance with one embodiment of the present invention. Referring to the Figure, the transmitter unit **102** in one embodiment includes an analog interface **201** configured to communicate with the sensor **101** (FIG. 1), a user input **202**, and a temperature measurement section **203**, each of which is operatively coupled to a transmitter processor **204** such as a central processing unit (CPU). As can be seen from FIG. 2, there are provided four contacts, three of which are electrodes—work electrode (W) **210**, guard contact (G) **211**, reference electrode (R) **212**, and counter electrode (C) **213**, each operatively coupled to the analog interface **201** of the transmitter unit **102** for connection to the sensor **101** (FIG. 1). In one embodiment, each of the work electrode (W) **210**, guard contact (G) **211**, reference electrode (R) **212**, and counter electrode (C) **213** may be made using a conductive material that is either printed or etched or ablated, for example, such as carbon which may be printed, or a metal such as a metal foil (e.g., gold) or the like, which may be etched or ablated or otherwise processed to provide one or more electrodes. Fewer or greater electrodes and/or contact may be provided in certain embodiments.

Further shown in FIG. 2 are a transmitter serial communication section **205** and an RF transmitter **206**, each of which is also operatively coupled to the transmitter processor **204**. Moreover, a power supply **207** such as a battery is also provided in the transmitter unit **102** to provide the necessary power for the transmitter unit **102**. Additionally, as can be seen from the Figure, clock **208** is provided to, among others, supply real time information to the transmitter processor **204**.

In one embodiment, a unidirectional input path is established from the sensor **101** (FIG. 1) and/or manufacturing and testing equipment to the analog interface **201** of the transmitter unit **102**, while a unidirectional output is established from the output of the RF transmitter **206** of the transmitter unit **102** for transmission to the primary receiver unit **104**. In this manner, a data path is shown in FIG. 2 between the aforementioned unidirectional input and output via a dedicated link **209** from the analog interface **201** to serial communication section **205**, thereafter to the processor **204**, and then to the RF transmitter **206**. As such, in one embodiment, via the data path described above, the transmitter unit **102** is configured to transmit to the primary receiver unit **104** (FIG. 1), via the communication link **103** (FIG. 1), processed and encoded data signals received from the sensor **101** (FIG. 1). Additionally, the unidirectional communication data path between the analog interface **201** and the RF transmitter **206** discussed above allows for the configuration of the transmitter unit **102** for operation upon completion of the manufacturing process as well as for direct communication for diagnostic and testing purposes.

As discussed above, the transmitter processor **204** is configured to transmit control signals to the various sections of the transmitter unit **102** during the operation of the transmitter unit **102**. In one embodiment, the transmitter processor **204** also includes a memory (not shown) for storing data such as the identification information for the transmitter unit **102**, as well as the data signals received from the sensor **101**. The stored information may be retrieved and processed for transmission to the primary receiver unit **104** under the control of the transmitter processor **204**. Furthermore, the power supply **207** may include a commercially available battery, which may be a rechargeable battery.

In certain embodiments, the transmitter unit **102** is also configured such that the power supply section **207** is capable of providing power to the transmitter for a minimum of about three months of continuous operation, e.g., after having been stored for about eighteen months such as stored in a low-power (non-operating) mode. In one embodiment, this may be achieved by the transmitter processor **204** operating in low power modes in the non-operating state, for example, drawing no more than approximately 1 μ A of current. Indeed, in one embodiment, a step during the manufacturing process of the transmitter unit **102** may place the transmitter unit **102** in the lower power, non-operating state (i.e., post-manufacture sleep mode). In this manner, the shelf life of the transmitter unit **102** may be significantly improved. Moreover, as shown in FIG. 2, while the power supply unit **207** is shown as coupled to the processor **204**, and as such, the processor **204** is configured to provide control of the power supply unit **207**, it should be noted that within the scope of the present invention, the power supply unit **207** is configured to provide the necessary power to each of the components of the transmitter unit **102** shown in FIG. 2.

Referring back to FIG. 2, the power supply section **207** of the transmitter unit **102** in one embodiment may include a rechargeable battery unit that may be recharged by a separate power supply recharging unit (for example, provided in the receiver unit **104**) so that the transmitter unit **102** may be powered for a longer period of usage time. Moreover, in one embodiment, the transmitter unit **102** may be configured without a battery in the power supply section **207**, in which case the transmitter unit **102** may be configured to receive power from an external power supply source (for example, a battery) as discussed in further detail below.

Referring yet again to FIG. 2, the temperature measurement section **203** of the transmitter unit **102** is configured to monitor the temperature of the skin near the sensor insertion site. The temperature reading is used to adjust the analyte readings obtained from the analog interface **201**. In certain embodiments, the RF transmitter **206** of the transmitter unit **102** may be configured for operation in the frequency band of approximately 315 MHz to approximately 322 MHz, for example, in the United States. In certain embodiments, the RF transmitter **206** of the transmitter unit **102** may be configured for operation in the frequency band of approximately 400 MHz to approximately 470 MHz. Further, in one embodiment, the RF transmitter **206** is configured to modulate the carrier frequency by performing Frequency Shift Keying and Manchester encoding. In one embodiment, the data transmission rate is about 19,200 symbols per second, with a minimum transmission range for communication with the primary receiver unit **104**.

Referring yet again to FIG. 2, also shown is a leak detection circuit **214** coupled to the guard contact (G) **211** and the processor **204** in the transmitter unit **102** of the data monitoring and management system **100**. The leak detection circuit **214** in accordance with one embodiment of the present invention may be configured to detect leakage current in the sensor **101** to determine whether the measured sensor data are corrupt or whether the measured data from the sensor **101** is accurate. Exemplary analyte systems that may be employed are described in, for example, U.S. Pat. Nos. 6,134,461, 6,175,752, 6,121,611, 6,560,471, 6,746,582, and elsewhere, the disclosure of each of which are incorporated by reference for all purposes.

FIG. 3 is a block diagram of the receiver/monitor unit of the data monitoring and management system shown in FIG. 1 in accordance with one embodiment of the present inven-

tion. Referring to FIG. 3, the primary receiver unit **104** includes an analyte test strip, e.g., blood glucose test strip, interface **301**, an RF receiver **302**, an input **303**, a temperature monitor section **304**, and a clock **305**, each of which is operatively coupled to a receiver processor **307**. As can be further seen from the Figure, the primary receiver unit **104** also includes a power supply **306** operatively coupled to a power conversion and monitoring section **308**. Further, the power conversion and monitoring section **308** is also coupled to the receiver processor **307**. Moreover, also shown are a receiver serial communication section **309**, and an output **310**, each operatively coupled to the receiver processor **307**.

In one embodiment, the test strip interface **301** includes a glucose level testing portion to receive a manual insertion of a glucose test strip, and thereby determine and display the glucose level of the test strip on the output **310** of the primary receiver unit **104**. This manual testing of glucose may be used to calibrate the sensor **101** or otherwise. The RF receiver **302** is configured to communicate, via the communication link **103** (FIG. 1) with the RF transmitter **206** of the transmitter unit **102**, to receive encoded data signals from the transmitter unit **102** for, among others, signal mixing, demodulation, and other data processing. The input **303** of the primary receiver unit **104** is configured to allow the user to enter information into the primary receiver unit **104** as needed. In one aspect, the input **303** may include one or more keys of a keypad, a touch-sensitive screen, or a voice-activated input command unit. The temperature monitor section **304** is configured to provide temperature information of the primary receiver unit **104** to the receiver processor **307**, while the clock **305** provides, among others, real time information to the receiver processor **307**.

Each of the various components of the primary receiver unit **104** shown in FIG. 3 is powered by the power supply **306** which, in one embodiment, includes a battery. Furthermore, the power conversion and monitoring section **308** is configured to monitor the power usage by the various components in the primary receiver unit **104** for effective power management and to alert the user, for example, in the event of power usage which renders the primary receiver unit **104** in sub-optimal operating conditions. An example of such sub-optimal operating condition may include, for example, operating the vibration output mode (as discussed below) for a period of time thus substantially draining the power supply **306** while the processor **307** (thus, the primary receiver unit **104**) is turned on. Moreover, the power conversion and monitoring section **308** may additionally be configured to include a reverse polarity protection circuit such as a field effect transistor (FET) configured as a battery activated switch.

The serial communication section **309** in the primary receiver unit **104** is configured to provide a bi-directional communication path from the testing and/or manufacturing equipment for, among others, initialization, testing, and configuration of the primary receiver unit **104**. Serial communication section **309** can also be used to upload data to a computer, such as time-stamped blood glucose data. The communication link with an external device (not shown) can be made, for example, by cable, infrared (IR) or RF link. The output **310** of the primary receiver unit **104** is configured to provide, among others, a graphical user interface (GUI) such as a liquid crystal display (LCD) for displaying information. Additionally, the output **310** may also include an integrated speaker for outputting audible signals as well as to provide vibration output as commonly found in handheld electronic devices, such as mobile telephones presently available. In a

further embodiment, the primary receiver unit **104** also includes an electro-luminescent lamp configured to provide backlighting to the output **310** for output visual display in dark ambient surroundings.

Referring back to FIG. 3, the primary receiver unit **104** in one embodiment may also include a storage section such as a programmable, non-volatile memory device as part of the processor **307**, or provided separately in the primary receiver unit **104**, operatively coupled to the processor **307**. The processor **307** may be configured to synchronize with a transmitter, e.g., using Manchester decoding or the like, as well as error detection and correction upon the encoded data signals received from the transmitter unit **102** via the communication link **103**.

Additional description of the RF communication between the transmitter **102** and the primary receiver **104** (or with the secondary receiver **106**) that may be employed in embodiments of the subject invention is disclosed in U.S. patent application Ser. No. 11/060,365 filed Feb. 16, 2005, now U.S. Pat. No. 8,771,183, entitled "Method and System for Providing Data Communication in Continuous Glucose Monitoring and Management System" the disclosure of which is incorporated herein by reference for all purposes.

Referring to the Figures, in one embodiment, the transmitter **102** (FIG. 1) may be configured to generate data packets for periodic transmission to one or more of the receiver units **104**, **106**, where each data packet includes in one embodiment two categories of data—urgent data and non-urgent data. For example, urgent data such as for example glucose data from the sensor and/or temperature data associated with the sensor may be packed in each data packet in addition to non-urgent data, where the non-urgent data is rolled or varied with each data packet transmission.

That is, the non-urgent data is transmitted at a timed interval so as to maintain the integrity of the analyte monitoring system without being transmitted over the RF communication link with each data transmission packet from the transmitter **102**. In this manner, the non-urgent data, for example that are not time sensitive, may be periodically transmitted (and not with each data packet transmission) or broken up into predetermined number of segments and sent or transmitted over multiple packets, while the urgent data is transmitted substantially in its entirety with each data transmission.

Referring again to the Figures, upon receiving the data packets from the transmitter **102**, the one or more receiver units **104**, **106** may be configured to parse the received data packet to separate the urgent data from the non-urgent data, and also, may be configured to store the urgent data and the non-urgent data, e.g., in a hierarchical manner. In accordance with the particular configuration of the data packet or the data transmission protocol, more or less data may be transmitted as part of the urgent data, or the non-urgent rolling data. That is, within the scope of the present disclosure, the specific data packet implementation such as the number of bits per packet, and the like, may vary based on, among others, the communication protocol, data transmission time window, and so on.

In an exemplary embodiment, different types of data packets may be identified accordingly. For example, identification in certain exemplary embodiments may include—(1) single sensor, one minute of data, (2) two or multiple sensors, (3) dual sensor, alternate one minute data, and (4) response packet. For single sensor one minute data packet, in one embodiment, the transmitter **102** may be configured to generate the data packet in the manner, or similar to the manner, shown in Table 1 below.

TABLE 1

Single sensor, one minute of data	
Number of Bits	Data Field
8	Transmit Time
14	Sensor1 Current Data
14	Sensor1 Historic Data
8	Transmit Status
12	AUX Counter
12	AUX Thermistor 1
12	AUX Thermistor 2
8	Rolling-Data-1

As shown in Table 1 above, the transmitter data packet in one embodiment may include 8 bits of transmit time data, 14 bits of current sensor data, 14 bits of preceding sensor data, 8 bits of transmitter status data, 12 bits of auxiliary counter data, 12 bits of auxiliary thermistor 1 data, 12 bits of auxiliary thermistor 2 data and 8 bits of rolling data. In one embodiment of the present invention, the data packet generated by the transmitter for transmission over the RF communication link may include all or some of the data shown above in Table 1.

Referring back, the 14 bits of the current sensor data provides the real time or current sensor data associated with the detected analyte level, while the 14 bits of the sensor historic or preceding sensor data includes the sensor data associated with the detected analyte level one minute ago. In this manner, in the case where the receiver unit **104**, **106** drops or fails to successfully receive the data packet from the transmitter **102** in the minute by minute transmission, the receiver unit **104**, **106** may be able to capture the sensor data of a prior minute transmission from a subsequent minute transmission.

Referring again to Table 1, the Auxiliary data in one embodiment may include one or more of the patient's skin temperature data, a temperature gradient data, reference data, and counter electrode voltage. The transmitter status field may include status data that is configured to indicate corrupt data for the current transmission (for example, if shown as BAD status (as opposed to GOOD status which indicates that the data in the current transmission is not corrupt)). Furthermore, the rolling data field is configured to include the non-urgent data, and in one embodiment, may be associated with the time-hop sequence number. In addition, the Transmitter Time field in one embodiment includes a protocol value that is configured to start at zero and is incremented by one with each data packet. In one aspect, the transmitter time data may be used to synchronize the data transmission window with the receiver unit **104**, **106**, and also, provide an index for the Rolling data field.

In a further embodiment, the transmitter data packet may be configured to provide or transmit analyte sensor data from two or more independent analyte sensors. The sensors may relate to the same or different analyte or property. In such a case, the data packet from the transmitter **102** may be configured to include 14 bits of the current sensor data from both sensors in the embodiment in which 2 sensors are employed. In this case, the data packet does not include the immediately preceding sensor data in the current data packet transmission. Instead, a second analyte sensor data is transmitted with a first analyte sensor data.

11

TABLE 2

Dual sensor data	
Number of Bits	Data Field
8	Transmit Time
14	Sensor1 Current Data
14	Sensor2 Current Data
8	Transmit Status
12	AUX Counter
12	AUX Thermistor 1
12	AUX Thermistor 2
8	Rolling-Data-1

In a further embodiment, the transmitter data packet may be alternated with each transmission between two analyte sensors, for example, alternating between the data packet shown in Table 3 and Table 4 below.

TABLE 3

Sensor Data Packet Alternate 1	
Number of Bits	Data Field
8	Transmitter Time
14	Sensor1 Current Data
14	Sensor1 Historic Data
8	Transmit Status
12	AUX Counter
12	AUX Thermistor 1
12	AUX Thermistor 2
8	Rolling-Data-1

TABLE 4

Sensor Data Packet Alternate 2	
Number of Bits	Data Field
8	Transmitter Time
14	Sensor1 Current Data
14	Sensor2 Current Data
8	Transmit Status
12	AUX Counter
12	AUX Thermistor 1
12	AUX Thermistor 2
8	Rolling-Data-1

As shown above in reference to Tables 3 and 4, the minute by minute data packet transmission from the transmitter **102** (FIG. 1) in one embodiment may alternate between the data packet shown in Table 3 and the data packet shown in Table 4. More specifically, the transmitter **102** may be configured in one embodiment transmit the current sensor data of the first sensor and the preceding sensor data of the first sensor (Table 3), as well as the rolling data, and further, at the subsequent transmission, the transmitter **102** may be configured to transmit the current sensor data of the first and the second sensor in addition to the rolling data (Table 4).

In one embodiment, the rolling data transmitted with each data packet may include a sequence of various predetermined types of data that are considered not-urgent or not time sensitive. That is, in one embodiment, the following list of data shown in Table 5 may be sequentially included in the 8 bits of transmitter data packet, and not transmitted with each data packet transmission of the transmitter (for example, with each 60 second data transmission from the transmitter **102**).

12

TABLE 5

Rolling Data		
Time Slot	Bits	Rolling-Data
0	8	Mode
1	8	Glucose1 Slope
2	8	Glucose2 Slope
3	8	Ref-R
4	8	Hobbs Counter, Ref-R
5	8	Hobbs Counter
6	8	Hobbs Counter
7	8	Sensor Count

As can be seen from Table 5 above, in one embodiment, a sequence of rolling data are appended or added to the transmitter data packet with each data transmission time slot. In one embodiment, there may be 256 time slots for data transmission by the transmitter **102** (FIG. 1), and where, each time slot is separated by approximately 60 second interval. For example, referring to the Table 5 above, the data packet in transmission time slot 0 (zero) may include operational mode data (Mode) as the rolling data that is appended to the transmitted data packet. At the subsequent data transmission time slot (for example, approximately 60 seconds after the initial time slot (0)), the transmitted data packet may include the analyte sensor 1 calibration factor information (Glucose1 slope) as the rolling data. In this manner, with each data transmission, the rolling data may be updated over the 256 time slot cycle.

Referring again to Table 5, each rolling data field is described in further detail for various embodiments. For example, the Mode data may include information related to the different operating modes such as, but not limited to, the data packet type, the type of battery used, diagnostic routines, single sensor or multiple sensor input, type of data transmission (RF communication link or other data link such as serial connection). Further, the Glucose1-slope data may include an 8-bit scaling factor or calibration data for first sensor (scaling factor for sensor 1 data), while Glucose2-slope data may include an 8-bit scaling factor or calibration data for the second analyte sensor (in the embodiment including more than one analyte sensors).

In addition, the Ref-R data may include 12 bits of on-board reference resistor used to calibrate our temperature measurement in the thermistor circuit (where 8 bits are transmitted in time slot 3, and the remaining 4 bits are transmitted in time slot 4), and the 20-bit Hobbs counter data may be separately transmitted in three time slots (for example, in time slot 4, time slot 5 and time slot 6) to add up to 20 bits. In one embodiment, the Hobbs counter may be configured to count each occurrence of the data transmission (for example, a packet transmission at approximately 60 second intervals) and may be incremented by a count of one (1).

In one aspect, the Hobbs counter is stored in a nonvolatile memory of the transmitter unit **102** (FIG. 1) and may be used to ascertain the power supply status information such as, for example, the estimated battery life remaining in the transmitter unit **102**. That is, with each sensor replacement, the Hobbs counter is not reset, but rather, continues the count with each replacement of the sensor **101** to establish contact with the transmitter unit **102** such that, over an extended usage time period of the transmitter unit **102**, it may be possible to determine, based on the Hobbs count information, the amount of consumed battery life in the transmitter unit **102**, and also, an estimated remaining life of the battery in the transmitter unit **102**.

13

That is, in one embodiment, the 20 bit Hobbs counter is incremented by one each time the transmitter unit **102** transmits a data packet (for example, approximately each 60 seconds), and based on the count information in the Hobbs counter, in one aspect, the battery life of the transmitter unit **102** may be estimated. In this manner, in configurations of the transmitter unit **620** (see FIG. 6) where the power supply is not a replaceable component but rather, embedded within the housing the transmitter unit **620**, it is possible to estimate the remaining life of the embedded battery within the transmitter unit **620**. Moreover, the Hobbs counter is configured to remain persistent in the memory device of the transmitter unit **620** such that, even when the transmitter unit power is turned off or powered down (for example, during the periodic sensor replacement, RF transmission turned off period and the like), the Hobbs counter information is retained.

Referring to Table 5 above, the transmitted rolling data may also include 8 bits of sensor count information (for example, transmitted in time slot 7). The 8 bit sensor count is incremented by one each time a new sensor is connected to the transmitter unit. The ASIC configuration of the transmitter unit (or a microprocessor based transmitter configuration or with discrete components) may be configured to store in a nonvolatile memory unit the sensor count information and transmit it to the primary receiver unit **104** (for example). In turn, the primary receiver unit **104** (and/or the secondary receiver unit **106**) may be configured to determine whether it is receiving data from the transmitter unit that is associated with the same sensor (based on the sensor count information), or from a new or replaced sensor (which will have a sensor count incremented by one from the prior sensor count). In this manner, in one aspect, the receiver unit (primary or secondary) may be configured to prevent reuse of the same sensor by the user based on verifying the sensor count information associated with the data transmission received from the transmitter unit **102**. In addition, in a further aspect, user notification may be associated with one or more of these parameters. Further, the receiver unit (primary or secondary) may be configured to detect when a new sensor has been inserted, and thus prevent erroneous application of one or more calibration parameters determined in conjunction with a prior sensor, that may potentially result in false or inaccurate analyte level determination based on the sensor data.

FIG. 4 is a flowchart illustrating a data packet procedure including rolling data for transmission in accordance with one embodiment of the present invention. Referring to FIG. 4, in one embodiment, a counter is initialized (for example, to T=0) (**410**). Thereafter the associated rolling data is retrieved from memory device, for example (**420**), and also, the time sensitive or urgent data is retrieved (**430**). In one embodiment, the retrieval of the rolling data (**420**) and the retrieval of the time sensitive data (**430**) may be retrieved at substantially the same time.

Referring back to FIG. 4, with the rolling data and the time sensitive data, for example, the data packet for transmission is generated (**440**), and upon transmission, the counter is incremented by one (**450**) and the routine returns to retrieval of the rolling data (**420**). In this manner, in one embodiment, the urgent time sensitive data as well as the non-urgent data may be incorporated in the same data packet and transmitted by the transmitter **102** (FIG. 1) to a remote device such as one or more of the receivers **104**, **106**. Furthermore, as discussed above, the rolling data may be

14

updated at a predetermined time interval which is longer than the time interval for each data packet transmission from the transmitter **102** (FIG. 1).

FIG. 5 is a flowchart illustrating data processing of the received data packet including the rolling data in accordance with one embodiment of the present invention. Referring to FIG. 5, when the data packet is received (**510**) (for example, by one or more of the receivers **104**, **106**, in one embodiment), the received data packet is parsed so that the urgent data may be separated from the not-urgent data (stored in, for example, the rolling data field in the data packet) (**520**). Thereafter the parsed data is suitably stored in an appropriate memory or storage device (**530**).

In the manner described above, in accordance with one embodiment of the present invention, there is provided method and apparatus for separating non-urgent type data (for example, data associated with calibration) from urgent type data (for example, monitored analyte related data) to be transmitted over the communication link to minimize the potential burden or constraint on the available transmission time. More specifically, in one embodiment, non-urgent data may be separated from data that is required by the communication system to be transmitted immediately, and transmitted over the communication link together while maintaining a minimum transmission time window. In one embodiment, the non-urgent data may be parsed or broken up in to a number of data segments, and transmitted over multiple data packets. The time sensitive immediate data (for example, the analyte sensor data, temperature data, etc.), may be transmitted over the communication link substantially in its entirety with each data packet or transmission.

FIG. 6 is a block diagram illustrating the sensor and the transmitter unit of the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention. Referring to FIG. 6, in one aspect, a transmitter unit **620** is provided in a substantially water tight and sealed housing. The transmitter unit **620** includes respective contacts (wrk, ref, cntr, and grd) for respectively establishing electrical contact with one or more of the working electrode, the reference electrode, the counter electrode and the ground terminal (or guard trace) of the sensor **610**. Also shown in FIG. 6 is a conductivity bar/trace **611** provided on the sensor **610**. For example, in one embodiment, the conductivity bar/trace **611** may comprise a carbon trace on a substrate layer of the sensor **610**. In this manner, in one embodiment, when the sensor **610** is coupled to the transmitter unit **620**, electrical contact is established, for example, via the conductivity bar/trace **611** between the contact pads or points of the transmitter unit **620** (for example, at the counter electrode contact (cntr) and the ground terminal contact (grd) such that the transmitter unit **620** may be powered for data communication.

That is, during manufacturing of the transmitter unit **620**, in one aspect, the transmitter unit **620** is configured to include a power supply such as battery **621**. Further, during the initial non-use period (e.g., post manufacturing sleep mode), the transmitter unit **620** is configured such that it is not used and thus drained by the components of the transmitter unit **620**. During the sleep mode, and prior to establishing electrical contact with the sensor **610** via the conductivity bar/trace **611**, the transmitter unit **620** is provided with a low power signal from, for example, a low power voltage comparator **622**, via an electronic switch **623** to maintain the low power state of, for example, the transmitter unit **620** components. Thereafter, upon connection with the sensor **610**, and establishing electrical contact via the conductivity bar/trace **611**, the embedded power supply **621** of

15

the transmitter unit **620** is activated or powered up so that some of all of the components of the transmitter unit **620** are configured to receive the necessary power signals for operations related to, for example, data communication, processing and/or storage.

In one aspect, since the transmitter unit **620** is configured to a sealed housing without a separate replaceable battery compartment, in this manner, the power supply of the battery **621** is preserved during the post manufacturing sleep mode prior to use.

In a further aspect, the transmitter unit **620** may be disposed or positioned on a separate on-body mounting unit that may include, for example, an adhesive layer (on its bottom surface) to firmly retain the mounting unit on the skin of the user, and which is configured to receive or firmly position the transmitter unit **620** on the mounting unit during use. In one aspect, the mounting unit may be configured to at least partially retain the position of the sensor **610** in a transcutaneous manner so that at least a portion of the sensor is in fluid contact with the analyte of the user. Example embodiments of the mounting or base unit and its cooperation or coupling with the transmitter unit are provided, for example, in U.S. Pat. No. 6,175,752, incorporated herein by reference for all purposes.

In such a configuration, the power supply for the transmitter unit **620** may be provided within the housing of the mounting unit such that, the transmitter unit **620** may be configured to be powered on or activated upon placement of the transmitter unit **620** on the mounting unit and in electrical contact with the sensor **610**. For example, the sensor **610** may be provided pre-configured or integrated with the mounting unit and the insertion device such that, the user may position the sensor **610** on the skin layer of the user using the insertion device coupled to the mounting unit. Thereafter, upon transcutaneous positioning of the sensor **610**, the insertion device may be discarded or removed from the mounting unit, leaving behind the transcutaneously positioned sensor **610** and the mounting unit on the skin surface of the user.

Thereafter, when the transmitter unit **620** is positioned on, over or within the mounting unit, the battery or power supply provided within the mounting unit is configured to electrically couple to the transmitter unit **620** and/or the sensor **610**. Given that the sensor **610** and the mounting unit are provided as replaceable components for replacement every 3, 5, 7 days or other predetermined time periods, the user is conveniently not burdened with verifying the status of the power supply providing power to the transmitter unit **620** during use. That is, with the power supply or battery replaced with each replacement of the sensor **610**, a new power supply or battery will be provided with the new mounting unit for use with the transmitter unit **620**.

Referring to FIG. 6 again, in one aspect, when the sensor **610** is removed from the transmitter unit **620** (or vice versa), the electrical contact is broken and the conductivity bar/trace **611** returns to an open circuit. In this case, the transmitter unit **620** may be configured, to detect such condition and generate a last gasp transmission sent to the primary receiver unit **104** (and/or the secondary receiver unit **106**) indicating that the sensor **610** is disconnected from the transmitter unit **620**, and that the transmitter unit **620** is entering a powered down (or low power off) state. And the transmitter unit **620** is powered down into the sleep mode since the connection to the power supply (that is embedded within the transmitter unit **620** housing) is broken.

In this manner, in one aspect, the processor **624** of the transmitter unit **620** may be configured to generate the

16

appropriate one or more data or signals associated with the detection of sensor **610** disconnection for transmission to the receiver unit **104** (FIG. 1), and also, to initiate the power down procedure of the transmitter unit **620**. In one aspect, the components of the transmitter unit **620** may be configured to include application specific integrated circuit (ASIC) design with one or more state machines and one or more nonvolatile and/or volatile memory units such as, for example, EEPROMs and the like.

Referring again to FIGS. 1 and 6, in one embodiment, the communication between the transmitter unit **620** (or **102** of FIG. 1) and the primary receiver unit **104** (and/or the secondary receiver unit **106**) may be based on close proximity communication where bi-directional (or uni-directional) wireless communication is established when the devices are physically located in close proximity to each other. That is, in one embodiment, the transmitter unit **620** may be configured to receive very short range commands from the primary receiver unit **104** (FIG. 1) and perform one or more specific operations based on the received commands from the receiver unit **104**.

In one embodiment, to maintain secure communication between the transmitter unit and the data receiver unit, the transmitter unit ASIC may be configured to generate a unique close proximity key at power on or initialization. In one aspect, the 4 or 8 bit key may be generated based on, for example, the transmitter unit identification information, and which may be used to prevent undesirable or unintended communication. In a further aspect, the close proximity key may be generated by the receiver unit based on, for example, the transmitter identification information received by the transmitter unit during the initial synchronization or pairing procedure of the transmitter and the receiver units.

Referring again to FIGS. 1 and 6, in one embodiment, the transmitter unit ASIC configuration may include a 32 KHz oscillator and a counter which may be configured to drive the state machine in the transmitter unit ASIC. The transmitter ASIC configuration may include a plurality of close proximity communication commands including, for example, new sensor initiation, pairing with the receiver unit, and RF communication control, among others. For example, when a new sensor is positioned and coupled to the transmitter unit so that the transmitter unit is powered on, the transmitter unit is configured to detect or receive a command from the receiver unit positioned in close proximity to the transmitter unit. For example, the receiver unit may be positioned within a couple of inches of the on-body position of the transmitter unit, and when the user activates or initiates a command associated with the new sensor initiation from the receiver unit, the transmitter unit is configured to receive the command from the receiver and, in its response data packet, transmit, among others, its identification information back to the receiver unit.

In one embodiment, the initial sensor initiation command does not require the use of the close proximity key. However, other predefined or preconfigured close-proximity commands may be configured to require the use of the 8 bit key (or a key of a different number of bits). For example, in one embodiment, the receiver unit may be configured to transmit a RF on/off command to turn on/off the RF communication module or unit in the transmitter unit **102**. Such RF on/off command in one embodiment includes the close proximity key as part of the transmitted command for reception by the transmitter unit.

During the period that the RF communication module or unit is turned off based on the received close proximity command, the transmitter unit does not transmit any data,

17

including any glucose related data. In one embodiment, the glucose related data from the sensor which are not transmitted by the transmitter unit during the time period when the RF communication module or unit of the transmitter unit is turned off may be stored in a memory or storage unit of the transmitter unit for subsequent transmission to the receiver unit when the transmitter unit RF communication module or unit is turned back on based on the RF-on command from the receiver unit. In this manner, in one embodiment, the transmitter unit may be powered down (temporarily, for example, during air travel) without removing the transmitter unit from the on-body position.

FIG. 7 is a flowchart illustrating data communication using close proximity commands in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention. Referring to FIG. 7, the primary receiver unit 104 (FIG. 1) in one aspect may be configured to retrieve or generate a close proximity command (710) for transmission to the transmitter unit 102. To establish the transmission range (720), the primary receiver unit 104 may be positioned physically close to (that is, within a predetermined distance from) the transmitter unit 102. For example, the transmission range for the close proximity communication may be established at approximately one foot distance or less between the transmitter unit 102 and the primary receiver unit 104. When the transmitter unit 102 and the primary receiver unit 104 are within the transmission range, the close proximity command, upon initiation from the receiver unit 104 may be transmitted to the transmitter unit 102 (730).

Referring back to FIG. 7, in response to the transmitted close proximity command, a response data packet or other responsive communication may be received (740). In one aspect, the response data packet or other responsive communication may include identification information of the transmitter unit 102 transmitting the response data packet or other response communication to the receiver unit 104. In one aspect, the receiver unit 104 may be configured to generate a key (for example, an 8 bit key or a key of a predetermined length) based on the transmitter identification information (750), and which may be used in subsequent close proximity communication between the transmitter unit 102 and the receiver unit 104.

In one aspect, the data communication including the generated key may allow the recipient of the data communication to recognize the sender of the data communication and confirm that the sender of the data communication is the intended data sending device, and thus, including data which is desired or anticipated by the recipient of the data communication. In this manner, in one embodiment, one or more close proximity commands may be configured to include the generated key as part of the transmitted data packet. Moreover, the generated key may be based on the transmitter ID or other suitable unique information so that the receiver unit 104 may use such information for purposes of generating the unique key for the bi-directional communication between the devices.

While the description above includes generating the key based on the transmitter unit 102 identification information, within the scope of the present disclosure, the key may be generated based on one or more other information associated with the transmitter unit 102, and/or the receiver unit combination. In a further embodiment, the key may be encrypted and stored in a memory unit or storage device in the transmitter unit 102 for transmission to the receiver unit 104.

FIG. 8 is a flowchart illustrating sensor insertion detection routine in the data monitoring and management system of

18

FIG. 1 in accordance with one embodiment of the present invention. Referring to FIG. 8, connection to an analyte sensor is detected (810), based on for example, a power up procedure where the sensor conduction trace 611 (FIG. 6) is configured to establish electrical contact with a predetermined one or more contact points on the transmitter unit 102. That is, when the sensor 101 (for example, the electrodes of the sensor) is correspondingly connected to the contact points on the transmitter unit 102, the transmitter unit 102 is configured to close the circuit connecting its power supply (for example, the battery 621 (FIG. 6)) to the components of the transmitter unit 102 and thereby exiting the power down or low power state into active or power up state.

In this manner, as discussed above, in one aspect, the transmitter unit 102 may be configured to include a power supply such as a battery 621 integrally provided within the sealed housing of the transmitter unit 102. When the transmitter unit 102 is connected or coupled to the respective electrodes of the analyte sensor that is positioned in a transcutaneous manner under the skin layer of the patient, the transmitter unit 102 is configured to wake up from its low power or sleep state (820), and power up the various components of the transmitter unit 102. In the active state, the transmitter unit 102 may be further configured to receive and process sensor signals received from the analyte sensor 101 (FIG. 1) (830), and thereafter, transmit the processed sensor signals (840) to, for example, the receiver unit 104 (FIG. 1).

Accordingly, in one aspect, the sensor 610 (FIG. 6) may be provided with a conduction trace 611 which may be used to wake up or exit the transmitter unit from its post manufacturing sleep mode into an active state, by for example, establishing a closed circuit with the power supply provided within the transmitter unit 102.

FIG. 9 is a flowchart illustrating sensor removal detection routine in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention. Referring to FIG. 9, when the sensor removal is detected (910) for example, based on detaching or removing the transmitter unit 102 that was in contact with the sensor 101, one or more status signal is generated (920), that includes, for example, an indication that the sensor removal state has been detected, and/or an indication that the transmitter unit 102 will enter a sleep mode or a powered down status. Thereafter, the generated status signal in one aspect is transmitted, for example, to the receiver unit 104 (930), and the transmitter unit 102 is configured to enter the power down mode or low power sleep mode (940).

In this manner, in one aspect, when the transmitter unit 102 is disconnected with an active sensor 101, the transmitter unit 102 is configured to notify the receiver unit 104 that the sensor 101 has been disconnected or otherwise, signals from the sensor 101 are no longer received by the transmitter unit 102. After transmitting the one or more signals to notify the receiver unit 104, the transmitter unit 102 in one embodiment is configured to enter sleep mode or low power state during which no data related to the monitored analyte level is transmitted to the receiver unit 104.

FIG. 10 is a flowchart illustrating the pairing or synchronization routine in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention. Referring to FIG. 10, in one embodiment, the transmitter unit 102 may be configured to receive a sensor initiate close proximity command (1010) from the receiver unit 104 positioned within the close transmission range. Based on the received sensor initiate command, the

transmitter unit identification information may be retrieved (for example, from a nonvolatile memory) and transmitted (1020) to the receiver unit 104 or the sender of the sensor initiate command.

Referring back to FIG. 10, a communication key optionally encrypted is received (1030) in one embodiment, and thereafter, sensor related data is transmitted with the communication key on a periodic basis such as, every 60 seconds, five minutes, or any suitable predetermined time intervals (1040).

Referring now to FIG. 11, a flowchart illustrating the pairing or synchronization routine in the data monitoring and management system of FIG. 1 in accordance with another embodiment of the present invention is shown. That is, in one aspect, FIG. 11 illustrates the pairing or synchronization routine from the receiver unit 104. Referring back to FIG. 11, the sensor initiate command is transmitted to the transmitter unit 102 (1110) when the receiver unit 104 is positioned within a close transmission range. Thereafter, in one aspect, the transmitter identification information is received (1120) for example, from the transmitter unit that received the sensor initiate command. Thereafter, a communication key (optionally encrypted) may be generated and transmitted (1130) to the transmitter unit.

In the manner described above, in one embodiment, a simplified pairing or synchronization between the transmitter unit 102 and the receiver unit 104 may be established using, for example, close proximity commands between the devices. As described above, in one aspect, upon pairing or synchronization, the transmitter unit 102 may be configured to periodically transmit analyte level information to the receiver unit for further processing.

FIG. 12 is a flowchart illustrating the power supply determination in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention. That is, in one embodiment, using a counter, the receiver unit 104 may be configured to determine the power supply level of the transmitter unit 102 battery so as to determine a suitable time for replacement of the power supply or the transmitter unit 102 itself. Referring to FIG. 12, periodic data transmission is detected (1210), and a corresponding count in the counter is incremented for example, by one with each detected data transmission (1220). In particular, a Hobbs counter may be used in the rolling data configuration described above to provide a count that is associated with the transmitter unit data transmission occurrence.

Referring to FIG. 12, the updated or incremented count stored in the Hobbs counter is periodically transmitted in the data packet from the transmitter unit 102 to the receiver unit 104 (1230). Moreover, the incremented or updated count may be stored (1240) in a persistent nonvolatile memory unit of the transmitter unit 102. Accordingly, based on the number of data transmission occurrences, the battery power supply level may be estimated, and in turn, which may provide an indication as to when the battery (and thus the transmitter unit in the embodiment where the power supply is manufactured to be embedded within the transmitter unit housing) needs to be replaced.

Moreover, in one aspect, the incremented count in the Hobbs counter is stored in a persistent nonvolatile memory such that, the counter is not reset or otherwise restarted with each sensor replacement.

FIG. 13 is a flowchart illustrating close proximity command for RF communication control in the data monitoring and management system of FIG. 1 in accordance with one embodiment of the present invention. Referring to FIG. 13,

a close proximity command associated with communication status, for example is received (1310). In one aspect, the command associated with the communication status may include, for example, a communication module turn on or turn off command for, for example, turning on or turning off the associated RF communication device of the transmitter unit 102. Referring to FIG. 13, the communication status is determined (1320), and thereafter, modified based on the received command (1330).

That is, in one aspect, using one or more close proximity commands, the receiver unit 104 may be configured to control the RF communication of the transmitter unit 102 to, for example, disable or turn off the RF communication functionality for a predetermined time period. This may be particularly useful when used in air travel or other locations such as hospital settings, where RF communication devices need to be disabled. In one aspect, the close proximity command may be used to either turn on or turn off the RF communication module of the transmitter unit 102, such that, when the receiver unit 104 is positioned in close proximity to the transmitter unit 102, and the RF command is transmitted, the transmitter unit 102 is configured, in one embodiment, to either turn off or turn on the RF communication capability of the transmitter unit 102.

FIG. 14 is a flowchart illustrating analyte sensor identification routine in accordance with one embodiment of the present invention. Referring to FIG. 14, periodically, sensor counter information is received (1410), for example included as rolling data discussed above. The received sensor counter information may be stored in one or more storage units such as a memory unit. When the sensor counter information is received, a stored sensor counter information is retrieved (1420), and the retrieved sensor counter information is compared with the received sensor counter information (1430). Based on the comparison between the retrieved sensor counter information and the received sensor counter information, one or more signal is generated and output (1440).

That is, in one aspect, the sensor counter in the transmitter unit 102 may be configured to increment by one with each new sensor replacement. Thus, in one aspect, the sensor counter information may be associated with a particular sensor from which monitored analyte level information is generated and transmitted to the receiver unit 104. Accordingly, in one embodiment, based on the sensor counter information, the receiver unit 104 may be configured to ensure that the analyte related data is generated and received from the correct analyte sensor transmitted from the transmitter unit 102.

An analyte monitoring system in one aspect includes a data processing unit, and a control unit in wireless communication with the data processing unit, the control unit configured to transmit one or more predefined close proximity commands to the data processing unit, where the data processing unit is configured to perform one or more predefined functions in response to the received one or more predefined close proximity commands.

The data processing unit may include a close proximity receiver coupled to an antenna for receiving the one or more predefined close proximity commands from the control unit.

In one aspect, the data processing unit may be configured to transmit one or more signals related to a monitored analyte level in response to the received one or more predefined close proximity commands.

Further, the communication of the one or more predefined close proximity commands may be performed when the data processing unit and the control unit are within a predeter-

21

mined distance from each other, where the predetermined distance may include a distance of less than one foot.

Also, there may be provided a memory unit, which may include an EEPROM or any other suitable type of nonvolatile or volatile memory, or combinations thereof

In one embodiment, the control unit may include an application specific integrated circuit (ASIC) configuration.

The analyte sensor may be coupled to the data processing unit, where the data processing unit may receive one or more signals from the analyte sensor, and where the control unit and the data processing unit may be in wireless communication using RF communication protocol.

A method in accordance with one embodiment includes positioning a controller unit within a transmission range for close proximity communication, transmitting one or more predefined close proximity commands, and receiving a response packet in response to the transmitted one or more predefined close proximity commands.

The method may include generating a communication key associated with the response packet, where the communication key may be generated based at least in part on a transmitter unit identification information, and further, where the transmitter unit identification information may be included in the response packet.

The transmission range in one embodiment may include a distance of less than one foot.

The method may also include storing one or more of the predefined close proximity commands or the response packet, where storing may include storing the commands or the response packet in a nonvolatile memory device.

The method may include receiving one or more signals from the analyte sensor, and further, where the transmitting and receiving include RF communication protocol.

Various other modifications and alterations in the structure and method of operation of this invention will be apparent to those skilled in the art without departing from the scope and spirit of the invention. Although the invention has been described in connection with specific preferred embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific embodiments. It is intended that the following claims define the scope of the present invention and that structures and methods within the scope of these claims and their equivalents be covered thereby.

What is claimed is:

1. A hand-held receiver unit of a glucose monitoring system, comprising:

an antenna for wirelessly communicating with a remote transmitter unit and a second receiver unit via respective Bluetooth communication links, the remote transmitter unit comprising a processor, a power supply, temperature measurement section and a memory operatively coupled to the processor, and being operatively coupled with a glucose sensor comprising a plurality of electrodes including a working electrode and a reference electrode, wherein the glucose sensor is configured to be positioned with a first portion of the glucose sensor above a skin layer of the user and a second portion of the glucose sensor implanted transcutaneously below the skin layer of the user and in contact with a bodily fluid of the user, wherein the glucose sensor is configured to generate data signals associated with a glucose level of the bodily fluid, wherein the remote transmitter unit is in a proximity of the receiver unit;

a processor, a display comprising a graphical user interface, a power supply, a clock, a user input, and a

22

memory operatively coupled to the processor for storing instructions which, when executed by the processor, cause the processor to:

receive a key associated with the remote transmitter unit, wherein the key enables the hand-held receiver unit to identify the remote transmitter unit;

initiate, using the antenna, communication with the remote transmitter unit via a first Bluetooth communication link after verifying the key;

access a communication key uniquely associated with the remote transmitter unit and configured to facilitate secure bi-directional communication between the hand-held receiver unit and the remote transmitter unit, wherein the communication key is configured to enable the hand-held receiver unit to recognize the remote transmitter unit as the originator of data communications and to confirm the remote transmitter unit is the expected originator of data communications;

receive, using the antenna via the first Bluetooth communication link and in association with the communication key, a plurality of communication packets from the remote transmitter unit on a periodic basis, each communication packet comprising a first data segment and a second data segment, wherein the first data segment comprises data indicative of the glucose level of the bodily fluid, and the second data segment comprises auxiliary information associated with the remote transmitter unit or the glucose sensor;

process the data indicative of the glucose level of the bodily fluid received from the remote transmitter unit to determine the glucose level of the bodily fluid for display;

output a numerical representation of the determined glucose level of the bodily fluid in the graphical user interface of the display; and

transmit data indicative of the glucose level of the bodily fluid to the second receiver unit via a second Bluetooth communication link, wherein the second receiver unit is embodied as a wrist watch device comprising a processor, a display comprising a graphical user interface, a power supply, a clock, a user input, and an antenna.

2. The receiver unit of claim 1, wherein the communication key is generated by the remote transmitter unit during initialization and stored in the memory of the remote transmitter unit after generation.

3. The receiver unit of claim 1, wherein the instructions, when executed by the processor, cause the processor to receive the communication key uniquely associated with the remote transmitter unit from the remote transmitter unit.

4. The receiver unit of claim 3, wherein the communication key is encrypted prior to storage in the memory of the remote transmitter unit.

5. The receiver unit of claim 1, wherein the receiver unit receives a communication packet from the remote transmitter unit every minute.

6. The receiver unit of claim 1, wherein the receiver unit receives a communication packet from the remote transmitter unit every 5 minutes.

7. The receiver unit of claim 1, wherein the communication key is generated by the receiver unit during synchronization with the remote transmitter unit and based on a remote transmitter unit identification.

8. The receiver unit of claim 1, wherein the auxiliary information comprises information corresponding to a remaining operable life of the remote transmitter unit.

23

9. The receiver unit of claim 1, wherein the auxiliary information comprises information corresponding to data transmission time.

10. The receiver unit of claim 1, wherein the auxiliary information comprises information corresponding to remote transmitter unit status. 5

11. The receiver unit of claim 1, wherein the auxiliary information comprises temperature data generated by the temperature measurement section.

12. The receiver unit of claim 1, wherein the data indicative of the glucose level of the bodily fluid comprises current values of the data and historical values of the data. 10

13. The receiver unit of claim 1, wherein the instructions, when executed by the processors:

receive, via the antenna and in association with the communication key, a second plurality of communication packets from the remote transmitter unit, each communication packet not comprising data indicative of the glucose level of the bodily fluid. 15

14. A glucose monitoring device of a glucose monitoring system, comprising: 20

a glucose sensor comprising a plurality of electrodes including a working electrode and a reference electrode, wherein the glucose sensor is configured to be positioned with a first portion of the glucose sensor above a skin layer of the user and a second portion of the glucose sensor implanted transcutaneously below the skin layer of the user and in contact with a bodily fluid of the user, wherein the glucose sensor is configured to generate data signals associated with a glucose level of the bodily fluid, 25 30

a transmitter unit operatively coupled with the glucose sensor, wherein the transmitter unit is configured to wirelessly communicate data signals indicative of the glucose level of the bodily fluid to a first receiver unit of the glucose monitoring system and a second receiver unit of the glucose monitoring system using a Bluetooth communication link, the transmitter unit comprising a processor, a power supply, a temperature measurement section operatively coupled to the processor, and a memory operatively coupled to the processor for storing instructions which, when executed by the processor, cause the processor to: 35 40

initiate communication with the first receiver unit of the glucose monitoring system via a first Bluetooth communication link based on a key associated with the glucose monitoring device, wherein the key enables the first receiver unit to identify the transmitter unit, wherein the first receiver unit is a hand-held communication device comprising a processor, a display comprising a graphical user interface, a power supply, a clock, a user input, and an antenna; 45 50

initiate communication with a second receiver unit of the glucose monitoring system via a second Bluetooth communication link based on the key associated with the glucose monitoring device, wherein the key enables the second receiver unit to identify the transmitter unit, wherein the second receiver unit is embodied as a wrist watch device comprising a processor, a display comprising a graphical user interface, a power supply, a clock, a user input, and an antenna; 55 60

access a communication key uniquely associated with the glucose monitoring device and configured to facilitate secure bi-directional communication between the transmitter unit and the first receiver unit and between the transmitter unit and the second receiver unit, wherein the communication key is configured to enable the first 65

24

receiver unit and the second receiver unit to recognize the transmitter unit as the originator of data communications and to confirm the transmitter unit is the expected originator of data communications;

transmit, to the first receiver unit and in association with the communication key, a plurality of communication packets on a periodic basis, each communication packet comprising a first data segment and a second data segment, wherein the first data segment comprises data indicative of the glucose level of the bodily fluid, and the second data segment comprises auxiliary information associated with the transmitter unit or glucose sensor, wherein, in response to receiving the plurality of communication packets, the first receiver unit is configured to output a numerical representation of the glucose level of the bodily fluid in the graphical user interface of the display of the first receiver unit; and

transmit, to the second receiver unit and in association with the communication key, a plurality of communication packets on a periodic basis, each communication packet comprising a first data segment and a second data segment, wherein the first data segment comprises data indicative of the glucose level of the bodily fluid, and the second data segment comprises auxiliary information associated with the transmitter unit or glucose sensor, wherein, in response to receiving the plurality of communication packets, the second receiver unit is configured to output a numerical representation of the glucose level of the bodily fluid in the graphical user interface of the display of the second receiver unit.

15. The glucose monitoring device of claim 14, wherein the instructions when executed by the processor, cause the processor to:

generate the communication key during an initialization process; and
store the communication key in the memory after generation.

16. The glucose monitoring device of claim 15, wherein the instructions when executed by the processor, cause the processor to:

encrypt the communication key prior to storage.

17. The glucose monitoring device of claim 14, wherein the instructions when executed by the processor, cause the processor to:

transmit a data packet of the plurality of data packets every minute.

18. The glucose monitoring device of claim 14, wherein the instructions when executed by the processor, cause the processor to:

transmit a data packet of the plurality of data packets every five minutes.

19. The glucose monitoring device of claim 14, wherein the auxiliary information comprises information corresponding to a remaining operable life of the remote transmitter unit.

20. The glucose monitoring device of claim 14, wherein the auxiliary information comprises information corresponding to data transmission time.

21. The glucose monitoring device of claim 14, wherein the auxiliary information comprises information corresponding to remote transmitter unit status.

22. The glucose monitoring device of claim 14, wherein the auxiliary information comprises temperature data generated by the temperature measurement section.

23. The glucose monitoring device of claim 14, wherein the data indicative of the glucose level of the bodily fluid comprises current values of the data and historical values of the data.

24. The glucose monitoring device of claim 14, wherein the glucose sensor comprises a multi-analyte sensor, and the multi-analyte sensor is further configured to generate data signals associated with levels of the bodily fluid of at least one analyte in addition to glucose.

25. The glucose monitoring device of claim 24, wherein the at least one analyte in addition to glucose comprises ketone.

26. The glucose monitoring device of claim 24, wherein the at least one analyte in addition to glucose comprises lactate.

27. The glucose monitoring device of claim 14, wherein the transmitter unit is operatively coupled with a second analyte sensor and the second analyte sensor is configured to generate data signals associated with an analyte level of a second analyte, and

wherein the second data segment further comprises information identifying the glucose sensor or second analyte sensor.

28. The glucose monitoring device of claim 14, wherein the temperature measurement section is configured to monitor a temperature of the skin layer near a sensor insertion site of the glucose sensor, and wherein the instructions, when executed by the processor, cause the processor to adjust the data signals indicative of the glucose level of the bodily fluid based on the monitored temperature.

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